

Preliminary III: Pulse Reflections & Impedance Matching

Connor Fricke (fricke.59@osu.edu)

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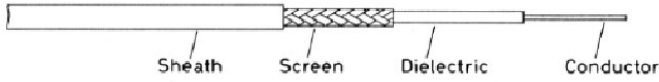


Fig. 1. Diagram of the typical construction of a coaxial cable. (See Ref. [1]).

I. INTRODUCTION

In physics experiments, it is very common to rely on the propagation of electrical signals. These signals are used to manipulate the instrumentation, but more importantly, they often carry information about the results of the experiment. In some contexts, the manipulation of these signals is not a trivial task, and some understanding of signal propagation is essential for obtaining accurate measurements with well calibrated instruments.

Most commonly, signals propagate through a special type of cable called a *coaxial cable*. In short, coax cables most often consist an outer plastic sheath, a woven conducting "shield", and an inner dielectric all surrounding the innermost conducting "core". The outer conductor is usually connected to ground while the inner conductor is driven to the signal voltage by a source at one end of the cable.

Like any electromagnetic waves, the signals propagating down a length of coaxial cable from a source to a measuring device are governed by Maxwell's equations. In general, at any interface between a cable and a device there will be a boundary which the signal must cross. At each boundary, some of the signal may be *reflected* while the rest of the signal is *transmitted*. Minimizing reflections is an important step to making accurate measurements of signals propagating via coax cable, as reflections can very quickly interfere with real signals, especially in high-frequency applications.

II. THEORY

One method of reducing reflections at a boundary is *impedance matching*. All cables have a *characteristic impedance*, Z , which is governed by the construction of the cable—the dielectric material between inner and outer conductors as well as their radii. By closely matching this characteristic impedance across a boundary, the reflections at the boundary can be minimized.

With a signal source at one end of the cable, and a load (such as a resistor) at the other end of the cable, the characteristic

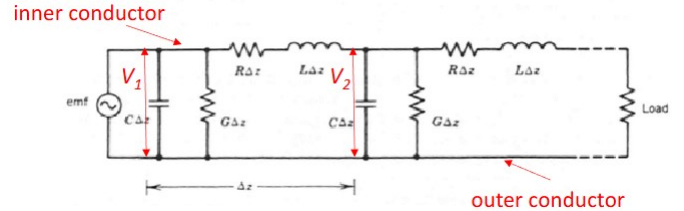


Fig. 2. Per unit length of coaxial cable, the construction can be thought of as the circuit above. A source of the signal is attached at the left end (emf), while a resistor is attached at the other end (load). An ideal cable will have small R and G . In a distributed circuit regime, V_1 and V_2 may be different. (See Refs. [1] and [2]).

impedance of the cable is approximated by

$$Z \approx \sqrt{\frac{L}{C}}$$

where L is the inductance per unit length of cable and C is the capacitance per unit length of cable (see Ref. [3]).

Deriving the reflection coefficient requires understanding Maxwell's equations at a boundary. At an interface between the coaxial transmission cable and the load, the tangential electric field and magnetic fields must be equivalent and continuous, satisfying

$$V_0 e^{i(\omega t - kx)} = \Gamma_r V_r e^{i(\omega t - kx)}$$

At this boundary, the amplitude of the reflection, V_r , is given by

$$V_r = \frac{R - Z}{R + Z} V_0 \quad (1)$$

where Z is the characteristic impedance of the cable and R is the resistance of the load. The rest of the power in the wave which is not reflected is transmitted to the load. Evidently, this gives a *reflection coefficient* of

$$\Gamma_r = \frac{V_r}{V_0} = \frac{R - Z}{R + Z} \quad (2)$$

Clearly, matching R as closely as possible to Z is the ideal scenario for minimizing the amplitude of waves reflected at the boundary.

III. ACTIVITY I: EXPERIMENT

The first part of this experiment involves the measurement of pulse reflections from the end of a cable terminated by a resistor of a known value. The experimental setup is shown in Figure 3.

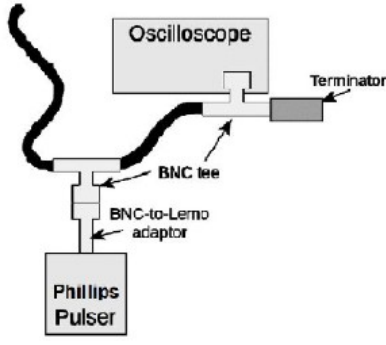


Fig. 3. Experimental setup showing the connection between the pocket pulser, the unterminated coaxial cable, and the oscilloscope used for measuring pulse and reflection amplitudes. The coaxial cable used in this experiment was about 20 feet of RG-58 cable with a characteristic impedance of $Z = 50 \Omega$.

At the loose (unconnected) end of the cable should be a printed circuit board with a resistor of a known value soldered in such a way that connects the core of the cable to the ground shielding via a PCB-to-SMA-to-BNC connection. This serves to provide the load (right side of Figure 2) which will need to be impedance matched.

A range of resistor values from $R = 10 \Omega$ to $R = 4.02 k\Omega$ as well as $R = 0$ (shorted) and $R = \infty$ (open) should be tested with this experimental setup. A bag of PCBs with different resistor values should be available, each of which can simply be screwed on to the SMA connection at the end of the coaxial cable. Each time a resistor is set in place, the oscilloscope should be used to measure the amplitude of the *reflected* pulse produced by the pocket pulser. Repeating the measurement of the amplitude several times allows for a standard deviation to be obtained for each measurement.

IV. ACTIVITY I: ANALYSIS

Once measurements of the reflection amplitude, V_r , for each resistor value, R , have been obtained, a plot of V_r against R can be generated (See Figure 4). A few weighted fits with fit parameters Z and V_0 produced the curves shown on the plot. A fit of equation (1) is shown in blue, most closely matching the data. Notably, there are still one or two recorded points which could be considered outliers as the line of best fit falls outside of one or even two standard deviations of the measurement. These outliers were observed for resistor values of 20Ω and 30Ω . Likely sources for the error could include a potentially faulty connection between the circuit board and the cable or imprecision with the oscilloscope cursors used to measure the amplitude of the reflection.

The second and third curve are weighted fits generated for equations (3) and (4), which are alternative equations given to us by friends,

$$V_r = \frac{R^2 - Z^2}{R^2 + Z^2} V_0 \quad (3)$$

and

$$V_r = \frac{\sqrt{R} - \sqrt{Z}}{\sqrt{R} + \sqrt{Z}} V_0 \quad (4)$$

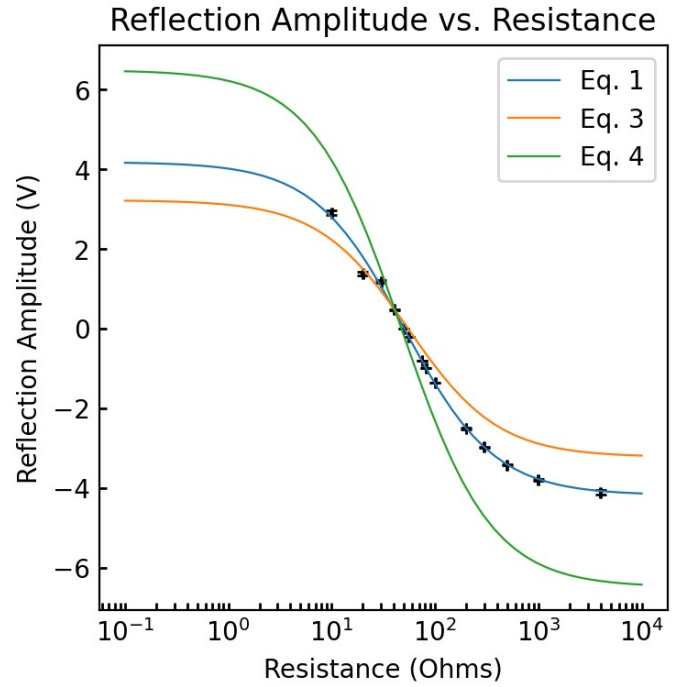


Fig. 4. Reflected pulse width plotted against PCB resistor value. Weighted fits for equations (1), (3) and (4) with $\tilde{\chi}^2 = 9.8$, 279.0, and 144.6 respectively suggesting equation (1) is the correct choice.

Based on visual inspection of the resulting fits, the original equation is clearly the correct one. The weighted fits produced the following table of results.

Fit Eq.	$Z (\Omega)$	V_0 (Volts)	$\tilde{\chi}^2$
(1)	50.4 ± 0.6	-4.18 ± 0.04	9.8
(3)	55 ± 3	-3.2 ± 0.1	279.0
(4)	47 ± 2	-6.5 ± 0.3	144.6

Evidently, equation (1) not only produced a relatively low $\tilde{\chi}^2$, but it also correctly predicted the characteristic impedance of the RG-58 cable with one standard deviation, which is provided by the manufacturer to have an expected value $Z = 50 \Omega$. Of course, this highlights the importance of impedance matching in minimizing reflections, and ultimately this is the reason for the terminator at the opposing end of the tee connected to the oscilloscope. It guarantees minimal reflections occur from the end of the tee thus any signal reflecting from the oscilloscope itself should be "damped" by this terminator.

V. ACTIVITY II: EXPERIMENT

Removing the resistor PCB from the end of the cable, we are left with the setup of Figure 3. This experiment aims to highlight the properties of reflections due to boundaries between cables of different characteristic impedance. The oscilloscope should be used to capture a waveform which displays both the pulse and the reflection in the form of PNG and CSV files. This waveform is shown in Figure 5.

After adding another 10 ft length of RG-58 cable (identical impedance), a waveform should be captured once again. The

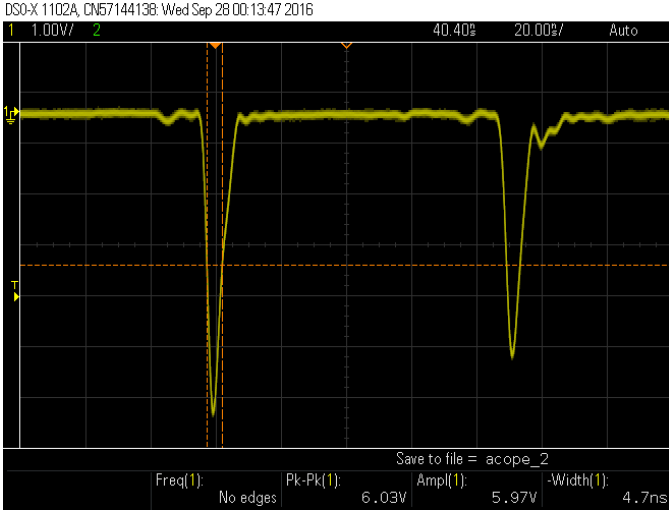


Fig. 5. Captured waveform of original pulse and reflected pulse with basic setup: no additional cable length after standard 20 ft of RG-58.

expectation is that the reflected pulse simply occurs at a later delay time. Because there is little to no variation in impedance at the interface between lengths of the identical cable, there should be little to no reflection from this interface, so expectations are still to see only two pulses.

Replacing the last RG-58 length of cable with another 10 ft length of RG-62 cable should produce not one, but two reflections, possibly more. We expect a reflection from RG-58 to RG-62 interface as well as a reflection from the end of the RG-62 cable. The initial reflection is due to the variation in impedance between the RG-58 ($Z = 50 \Omega$) and RG-62 ($Z = 93 \Omega$), and we expect it to occur at a delay time roughly equivalent to the time delay observed with no additional length of cable. The second pulse comes from the open end of the cable, and should occur slightly sooner than the pulse which reflected from the end of the 30 ft RG-58 cable. This is because the propagation velocity is higher in the RG-62 cable, at around $0.84c$ compared with the RG-58's propagation velocity of $0.76c$ (see Ref. [4]), thus the reflection returns slightly quicker through the RG-62 cable. Additional reflections from the RG-62 to RG-58 interface (the return trip for the cable-end reflection) may appear with short delays from the main cable-end reflection but with greatly reduced amplitudes.

VI. ACTIVITY II: ANALYSIS

Obviously, with matched impedance across the cable interface (RG-58 to RG-58), we expect to see only two pulses in total: the true pulse and the pulse reflection from the open ($R = \infty$) end of 30 ft cable. In the ideal scenario, the reflected wave's amplitude would match that of the original pulse, but natural attenuation of the signal as it propagates the extra length of the cable means that there will still be some noticeable pulse amplitude loss. This waveform is shown in Figure 6 in blue. The reflection occurs with a delay of roughly 60 ns as opposed to the expected delay time of 53 ns based on a pulse propagation velocity of $0.76c$ over 40 ft. The reflection amplitude is attenuated naturally because the coaxial cable is not ideal, so some amplitude is lost over every unit length.

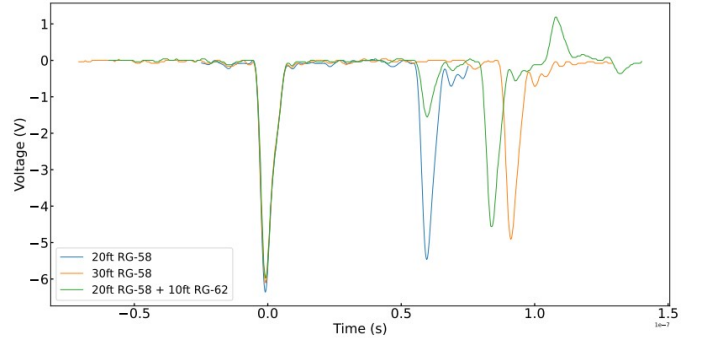


Fig. 6. Captured and plotted CSV data from all three setups: 20 ft of RG-58, 30 ft of RG-58, and 20 ft of RG-58 with 10 ft of RG-62. Pulse reflections seem to meet expectations as mentioned in the previous section.

In the case of 30 ft of RG-58 cable, we see the pulse reflection further delayed, by a total of roughly 92 ns as opposed to the expected delay time of around 80 ns over 60 ft of cable. Additionally, the reflection amplitude is now further decreased, consistent with natural attenuation over the additional 20 ft of propagation required by the reflection as compared to the previous setup.

The most interesting case here, as expected, is the setup with an additional 10 ft of RG-62 setup. As predicted, we see the reflection from the RG-58 to RG-62 interface appear at the same time as the main reflection from the basic setup (with no additional length). It has a greatly reduced amplitude as the reflection coefficient given by equation (2) is only $\Gamma_r \sim 0.3$ at this interface, meaning the reflection will be at most three tenths of the original amplitude without accounting for natural attenuation. The original pulse has an amplitude of 6V, while the reflection here is only around 1.5V, consistent with expectations. The secondary reflection (from the end of the RG-62 cable) occurs after the initial reflection, and only slightly sooner than the reflection for the 30 ft RG-58 case, consistent with the faster propagation velocity in the RG-62 cable.

The expected delay time of the reflection from the end of this cable is given by

$$\frac{40 \text{ ft}}{0.76c} + \frac{20 \text{ ft}}{0.84c} \approx 78 \text{ ns} \quad (5)$$

which is roughly consistent with the measured delay time of around 84 ns.

Additionally we see a few reflections of the reflection which show up with small amplitudes and additional delay as predicted in the previous section.

VII. CONCLUSIONS

Evidently, it is important to practice impedance matching, which includes using cables of identical characteristic impedance as well as the use of impedance-matched terminators on open ends of cables. These reflections can easily interfere with experimental measurements thus they must be eliminated. The line of best fit for activity I suggests that equation (1) is the correct choice for calculating the reflection coefficient, Γ_r . We were quite easily able to obtain the

characteristic impedance ($Z = 50 \, \Omega$) of the cable via fitting parameters of observed data. Of course, there are other factors at play such as natural attenuation per unit length, which is an unavoidable consequence of using non-ideal conductors in the lab.

REFERENCES

- [1] D. Preston and E. Dietz, *The Art of Experimental Physics*. John Wiley and Sons, Hoboken, New Jersey, 1991.
- [2] The Ohio State University, "Intro Lab III: Impedance Matching," *CarmenCanvas*, 2024.
- [3] T. H. Lee, *Planar Microwave Engineering: A Practical Guide to Theory, Measurement, and Circuits*. Cambridge University Press, 2004.
- [4] Belden, Inc, *8219 and 8254 Technical Data Sheets*, 2025.